t-ratios are found to be statistically significant, then multicollinearity is not a serious issue. One can also check whether the estimated coefficients are stable by deleting some observations (i. e., by changing sample size).

Check Your Progress 2

1) Examine whether the following statements are 'true' or 'false'. Give an explanation for your answer.

   a) Multicollinearity is a problem cannot be decided by just looking at the pair-wise correlation between the explanatory variables.

   b) High standard error of the estimate or of a variable in a regression equation has in an equation have, evidence of high multicollinearity.

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10.5 REMEDIAL MEASURES FOR MULTICOLLINEARITY

Some of the solutions to multicollinearity are suggested below:

1. Dropping one of the correlated variables

This method may be resorted to but it can cause omission bias. Moreover, if there is strong theoretical reason for including a variable, then exclusion of the variable is unacceptable. In the model $y = \beta_1 x_1 + \beta_2 x_2 + u$, suppose variables$x_1$ and x_2 are highly correlated. The OLS estimator of β_1 is denoted by $\hat{\beta}_1$. The mean and variance of $\hat{\beta}_1$ are such that:

$$E(\hat{\beta}_1) = \beta_1 \qquad \dots (10.24)$$

and

$$Var\ (\hat{\beta}_1) = \frac{\sigma^2}{\sum x_1{}^2(1-r_{12}{}^2)} \qquad \dots (10.25)$$

Now, we drop the variable x_2 such that the model looks like $y = \beta_1 x_1 + v$. The Omitted variable estimator of β_1 is denoted by $\widetilde{\beta_1}$.

$$\widetilde{\beta_1} = \frac{\sum x_1 y}{\sum x_1{}^2} = \frac{\sum x_1(\beta_1 x_1 + \beta_2 x_2 + \mu)}{\sum x_1{}^2}$$

$$\widetilde{\beta_1} = \beta_1 + \frac{S_{12}}{S_{11}}\beta_2 + \frac{\sum x_1 u}{S_{11}} \qquad \dots (10.26)$$

Here, $S_{12} = \sum x_1\, x_2, S_{11} = \sum x_1{}^2$

$$E(\widetilde{\beta_1}) = \beta_1 + \beta_2 \frac{S_{12}}{S_{11}} \qquad \dots (10.27)$$

Hence, $\widetilde{\beta_1}$ is biased.

$$Var(\beta_1^*) = Var(\frac{\sum x_1 u}{S_{11}}) = \frac{\sigma^2 S_{11}}{S_{11}^2} = \frac{\sigma^2}{S_{11}} \quad \ldots (10.28)$$

Comparing with the variance of $\hat{\beta}_1$, we see that $\widetilde{\beta_1}$ has a smaller variance. Greater the correlation coefficient r_{12}, smaller the variance of $\widetilde{\beta_1}$ as compared to $\hat{\beta}_1$.

$$\frac{Var(\widetilde{\beta_1})}{Var(\hat{\beta}_1)} = 1 - r_{12}^2 \quad \ldots (10.29)$$

Although $\widetilde{\beta_1}$ has lesser variance, it is biased. Omitting a variable worsens the problem of multicollinearity because it causes the estimators to be biased whereas with multicollinearity, the estimators are unbiased and continue to have the minimum variance among the linear unbiased estimators (BLUE).

2. Ignoring it

Even with multicollinearity, the OLS estimators are unbiased, efficient and consistent, that is, they continue to be BLUE. The problem of multicollinearity can be ignored if the model is nevertheless statistically adequate with the coefficients having appropriate sign. Sometimes, the presence of multicollinearity may not reduce t-ratios to such an extent as to render the variables insignificant. Moreover, multicollinearity is a sample or data problem over which the researcher has little control. Even if all the coefficients are not estimable, a linear combination of them may be estimable which serves well in many cases.

3. Ratio transformations of the correlated variables

The variables can be suitably transformed by expressing them in ratios, for instance, as a percentage of GDP or in per capita terms wherever applicable. However, ratio transformation of variables should be supported by theoretical logic. First difference transformation of variables can also be done. It is mainly done as a remedy for autocorrelation. These transformations reduce the collinearity between the variables. However, such transformations are not without problems. In taking first differences, one observation is lost, hence reducing degrees of freedom by one. This may affect the analysis if the sample size is small. Also, in ratio transformations, the error terms in the transformed model may become heteroscedastic if the original model has homoscedastic errors.

4. Expanding the sample size

You might have noticed by now that multicollinearity is more of a data problem. As we mentioned earlier, standard error can be increased by reducing the error variance or increasing $\sum x_{1i}^2$. Thus, collecting more data if possible or pooling the data across cross-sections and over time is another solution. The independent variables may not be collinear in the population although they may be found to be collinear in the given sample. For instance, theoretically, consumption is a function of both income and wealth. But, in a sample of observations, income and wealth may be found to be collinear as wealthy people may also be having higher incomes. This is especially true in time series data. In a cross-section of observations, it is possible to obtain a sample in which the relationship between wealth and income is random. Hence, if we have the 'right' sample, multicollinearity may not be reflected in the variables.

5. Using extraneous estimates

Let us consider the following equation:

$$\log Q = \rho + \beta logP + \theta logM + \varepsilon \qquad \ldots (10.30)$$

where Q = quantity, P = Price, $and\ M$ = income) in (10.30) estimating both price and income elasticity precisely is difficult in time series data as income and price are generally found to be correlated. In that case, estimate of income elasticity $\hat{\theta}$ can be obtained from budget studies and by regressing ($\log Q - \hat{\theta} logM$) on log P, estimates of ρ and β can be obtained. Here, $\hat{\theta}$ is called the extraneous estimate. However, the problem is that the cross-section estimate of θ as obtained from budget studies may not be the same as its time series estimate. Hence, this process is advised only when data is lacking. When both cross-section and time series data are available, pooling different data sets will enable getting more precise estimates.

6. Getting a priori information

Sometimes, the underlying economic theory may give relevant information. For example, let us consider the Cobb-Douglas production function,

$$Y = AK^{\alpha}L^{\beta} \qquad \ldots (10.31)$$

Where Y = output, K = capital, L = labour, A = efficiency parameter, α and β are share parameters of the production function. Here, K and L may be found to be collinear as the assumption of constant returns to (ERS)scale gives $\alpha + \beta = 1 implies\ \beta = 1 - \propto$). Now, we can express the above function in per labour terms as,

$$y = Ak^{\alpha} \qquad \frac{Y}{L} = \frac{AK^{\propto}L^{1-\propto}}{L}\ or\ y = A\,(K/L)^{\propto} = AK^{\propto} \qquad \ldots (10.32)$$

Regressing output per labour $'y'$ on capital per labour $'k'$ gives an estimate of α. But such a priori restriction(such as CRS) are generally supposed to be tested rather than imposed on the data.

7. Ridge Regression

Here, a constant **λ** is added to the variance of explanatory variables. In matrix notation, $\boldsymbol{X'X}$ is the variance-covariance matrix of explanatory variables. By adding **λ** to the diagonal elements of $\boldsymbol{X'X}$ matrix, the ridge estimator $\widehat{\boldsymbol{\beta}}_{\boldsymbol{R}}$ can be found. Doing so decreases the intercorrelation between explanatory variables.

$$\widehat{\boldsymbol{\beta}}_{\boldsymbol{R}} = (\boldsymbol{X'X} + \boldsymbol{\lambda}\mathbf{I})^{-1}\boldsymbol{X'y} \qquad \ldots (10.33)$$

Here, **X** is the matrix of explanatory variables and $\boldsymbol{y}$ is the vector of explained variables. The procedure of Ridge Regression is explained below.

(i) Constrained Least Squares

The residual sum of squares ($\boldsymbol{u'u}$) is minimised subject to the constraint that $\sum_{i=1}^{k} {\beta_i}^2$ *or* $\boldsymbol{\beta'\beta = c}$

Our objective function is which needs to be minimised,

$$(\boldsymbol{y - X\beta})'(\boldsymbol{y - X\beta}) + \boldsymbol{\lambda}(\boldsymbol{\beta'\beta - c}) \quad \ldots (10.34)$$

(Here, $\boldsymbol{\lambda}$ is the Lagrangian multiplier)

Setting the first order derivatives with respect to $\boldsymbol{\beta}$ equal to zero, we obtain

$$\boldsymbol{-2X'y + 2X'X\beta + 2\beta\lambda = 0}$$

Hence,

$$\boldsymbol{\hat{\beta}_R = (X'X + \lambda I)^{-1} X'y} \quad \ldots (10.35)$$

In the case of two explanatory variables, the constrained least squares minimisation problem is give as follows:

$$\text{Minimize } \sum (y - \beta_1 x_1 - \beta_2 x_2)^2 + \lambda\,({\beta_1}^2 + {\beta_2}^2 - c) \quad \ldots (10.36)$$

Setting the first order partial derivatives with respect to β_1 and β_2 equal to zero, we get,

$$2\sum (y - \beta_1 x_1 - \beta_2 x_2)(-x_1) + 2\lambda\,\beta_1 = 0 \quad \ldots (10.37)$$

$$2\sum (y - \beta_1 x_1 - \beta_2 x_2)(-x_2) + 2\lambda\,\beta_2 = 0 \quad \ldots (10.38)$$

Simplifying the above two equations, we get,

$$(S_{11} + \lambda)\beta_1 + S_{12}\beta_2 = S_{1y} \quad \ldots (10.39)$$

$$S_{12}\beta_1 + (S_{22} + \lambda)\beta_2 = S_{2y} \quad \ldots (10.40)$$

Here, $S_{11} = \sum {x_1}^2$; $S_{22} = \sum {x_2}^2$; $S_{12} = \sum x_1\, x_2\; s_{1y} = \sum x_1\, y$ *and* $s_{2y} = \sum x_2\, y$.

In (10.39) and (10.40) we have two equations in two unknowns (since date on x and y are available which β_1 *and* β_2 *are unknown*) Thus, we get the ridge regressionestimators for two explanatory variables and from these equations, λ can be estimated using the criterion ${\beta_1}^2 + {\beta_2}^2 = c$.

In the case described above, applying ridge regression is natural when prior information is given. But having prior information (in the above example that ${\beta_1}^2 + {\beta_2}^2 = c$) is not always possible. Hence, choosing λ is not always easy. You should note that λ is a function of β_i and σ^2 (variance of the error term), both of which are unknown. Some authors suggest that different values of λ may be tried and that value of λ may be chosen such that the coefficients do not have "unreasonable values". Others have suggested that in order to estimate λ, initial estimates of β_i and σ^2 can be used. Iterative procedure can be applied to get the iterated ridge estimator.

One shortcoming of ridge regression is that addition of λ to the variances of explanatory variables results in biased estimators but there is a decline in the mean squared error (MSE) as well. As pointed out by some authors, there always exists $\lambda > 0$ such that,

$\sum_{i=1}^{k} MSE\ (\tilde{\beta}_i) < \sum_{i=1}^{k} MSE\ (\hat{\beta}_i)$ … (10.41)

Here, k is the number of explanatory variables, $\tilde{\beta}_i$ is the ridge regression estimator of β_i and $\hat{\beta}_i$ is the OLS estimator of β_i. Another shortcoming is that ridge regression estimators differ when different linear transformations of independent variables are used. Also, ridge regression depends on units of measurement of the independent variables. If the explanatory variables are measured in different units, adding the same value of λ to the variance of different explanatory variables does not make sense. Normalization of variables however can solve this problem.

(ii) Bayesian Interpretation

Here, some prior information on regression parameters is combined with sample information. Assuming the prior information $\beta_i \sim$ IN $(0, \sigma_\beta^2)$, the ridge regression estimates of β_i's are obtained. Here, the ridge constant $\lambda = \sigma^2/\sigma_\beta^2$ where σ^2 has to be estimated as it is unknown. However, the mean of β_i's is equal to zero is not a reasonable assumption. Hence, the ridge estimator with Bayesian interpretation does not make much sense in econometrics. More complicated generalized ridge estimators are obtained by relaxing this assumption.

(iii) Measurement Error Interpretation

In this case, random errors with mean zero and variance 'λ' are added to the explanatory variables. Because the errors are random, the covariance between the explanatory variables is unaffected but the variance of explanatory variables increases by 'λ'. Hence, ridge regression estimator is obtained. In this method, the data is not precise as random errors are added but the estimates obtained by adopting such a technique are precise.

In conclusion, ridge regression is justified in some situations but involves subjective judgement or 'vague prior information' in others. Due to these shortcomings, it is not recommended to use ridge regression as a general solution to the problem of multicollinearity.

8. Principal Components Regression

Principal Component Analysis (PCA) is a multivariate statistical technique in which the 'k' correlated explanatory variables are used to derive uncorrelated indices or components such that each of these components is a linear function of the 'k' explanatory variables. Assuming 'k' explanatory variables ($x_1, x_2, ..., x_k$), the linear functions of these variables are as follows:

$z_1 = a_1x_1 + a_2x_2 + \cdots + a_kx_k$ … (10.42)

$z_2 = b_1x_1 + b_2x_2 + \cdots + b_kx_k$ … (10.43)

Where $z_1, z_2,...,z_k$ are the principal components. Hence, there are 'k' principal components. The first principal component 'z_1' is found out by choosing a's such that the variance of z_1 is maximised subject to the condition that:

$a_1^2 + a_2^2 + ... + a_k^2 = 1$(normalization condition) … (10.44)

Similarly, the second, third and subsequent principal components are derived. The first principal component 'z_1' has the highest variance.

$$Var(z_1) > Var(z_2) > Var(z_3) > .... > Var(z_k) \quad \dots (10.45)$$

The various principal components are orthogonal or perpendicular to each other, that is, they are not correlated to each other. Another property of the principal components is that:

$$Var(z_1) + Var(z_2) + Var(z_3) + ... + Var(z_k) = Var(x_1) + Var(x_2) + Var(x_3) + ... + Var(x_k) \quad \dots (10.46)$$

Now, how does the PCA help in multicollinearity? We can regress 'y' on the z's but regressing 'y' on z's and then substituting the value of z's in terms of x's amounts to the same thing as regressing 'y' on x's. Alternatively, one can regress 'y' on a few z's but there is no reason to believe that the first principal component having maximum variance would have the highest correlation with 'y' or that the order of principal components affects its correlation with 'y'. Another problem is that the principal components often do not have any economic logic. For instance, 2 (income) + 3 (price) makes no economic sense. Also, the units of measurement of x's affect the z's. Standardization of variables can solve this problem.

In many cases which involve the construction of indices, the PCA makes economic sense. The use of PCA can also give some prior information about restrictions on the parameters. Individual regression coefficients are sometimes not estimable but their linear combinations are. Having prior knowledge about parameter restrictions can help us in estimation of individual parameters.

9. Generalised Inverse Solution

There is a generalised inverse for every matrix **A** irrespective of its dimension. If **A** is a square matrix and it is invertible, that is, the determinant is non-zero, then $\mathbf{A^{-1}}$ can be easily found. But if **A** is a non-square matrix, say of order (m × n), or it is singular (determinant is zero), then the generalised inverse of **A** is indicated by **G** (n × m matrix) such that **AGA = A**. The generalised inverse of **A** is denoted by $\boldsymbol{A'}$.

$$\mathbf{AGA = A}\boldsymbol{A'}\mathbf{A = A} \quad \dots (10.47)$$

There are various generalised inverses G of an (m X n) matrix **A**. In the case of an invertible square matrix **A,** the generalised inverse G is the same as $\mathbf{A^{-1}}$. Hence $\mathbf{AA^{-1}A = A}$. For a non-singular matrix, the generalised inverse is unique.

For an (m X n) matrix having rank $r(A) = r$, if the matrix **A** can be partitioned such that $\mathbf{A} = \begin{pmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \boldsymbol{E} & \boldsymbol{F} \end{pmatrix}$, where **C** is ($r$ X r) non-singular matrix and $r(A) = r(C) = r$, then the generalised inverse of **A** is:

$$\mathbf{G} = \begin{pmatrix} \boldsymbol{C^{-1}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \quad \dots (10.48)$$

The procedure to compute **G** is as follows: First, take any (r X r) non-singular sub-matrix, that is, **C**. It is not necessary to take elements in **C** from adjacent rows and columns in A. Now, find $(C^{-1})'$ and replace C by $(C^{-1})'$. Remaining elements of **A** should be replaced by 0. Finally, transpose the resulting matrix.

Example 10.1

Let **A** = $\begin{pmatrix} 4 & 1 & 2 \\ 1 & 1 & 5 \\ 3 & 1 & 3 \end{pmatrix}$

You can find out that the determinant of the above matrix vanishes. Let us take (**A**) = 2. Accordingly, **C** = $\begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix}$.

Hence $\mathbf{C}^{-1} = \begin{pmatrix} 1/3 & -1/3 \\ -1/3 & 4/3 \end{pmatrix}$.

Thus, the generalised inverse, **G** = $\begin{pmatrix} 1/3 & -1/3 & 0 \\ -1/3 & 4/3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Using the generalised inverse, the regression parameter can be estimated as $\widehat{\boldsymbol{\beta}}_G = (\boldsymbol{X}'\boldsymbol{X})^{-}\boldsymbol{X}'\boldsymbol{y}$ for the model. Different solutions are obtained in parameter estimation by using different generalised inverses. Hence, a class of generalised inverse regression estimators can be obtained.

Check Your Progress 3

1) Explain the following methods:

   a) Ridge Regression

   b) Principal Component Regression

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2) A researcher estimates the following model for stock market returns, but thinks that there may be a problem with it. By calculating the *t*-ratios, and considering their significance (p-values given in brackets underneath) and by examining the value of R^2 or otherwise, suggest what the problem might be.

   $y_t = 0.638 + 0.402\ x_{2t} - 0.891\ x_{3t}$ $(R^2 = 0.96, \bar{R}^2 = 0.89)$

   (0.436) (0.291) (0.763)

   Figures in parentheses are the standard errors of the coefficients. Explain how you will proceed to solve the problem.

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3) "The relevant question to ask (if there is high multicollinearity) is not what variables to drop but what other information will help." True or False? Substantiate your answer.

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10.6 LET US SUM UP

Interpretation of individual regression coefficients becomes difficult due to high inter-correlation between explanatory variables. This is known as the multicollinearity problem. However, it must be noted that high inter-correlation between independent variables does not necessarily cause problems in parameter estimation and inference. Even with high inter-correlation, standard errors can be low if error variance is low and variation in independent variables is high. Measuring multicollinearity simply on the basis of inter-correlations is not advisable as these correlations can change by suitably transforming the independent variables. This, however, does not imply that the multicollinearity problem has been resolved. What is important is how precisely the coefficients or a linear combination of them are estimable. Solutions to multicollinearity such as ridge regression are ad hoc measures requiring a priori information. Essentially, the problem arises due to paucity of information. It is advisable to get more data (if possible) and to examine the prior information available.

10.7 KEY WORDS

Multicollinearity : It is basically, high inter-correlation between explanatory variables in a regression model. If the correlation between explanatory variables is found to be very high, it causes problems in parameter estimation by inflating the standard errors and making t-ratios statistically insignificant.

Tolerance (TOL) : The inverse of VIF is known as Tolerance (TOL) and it is another indicator of collinearity between variables.

Variance Inflation Factor (VIF) : It is an indicator of multicollinearity. In the regression model $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\mu}_i$, the VIF is given by $VIF = \frac{1}{(1-r_{12}{}^2)}$.

10.8 SOME USEFUL BOOKS/ REFERENCES

Baltagi, Badi H. (2008), '*Econometrics*,' Fourth Edition, Springer-Verlag Berlin, Heidelberg, ISBN 978-3-540-76515-8

Brooks, Chris (2014), '*Introductory Econometrics for Finance*,' Third Edition, Cambridge University Press

Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill

Maddala, G. S. and K. Lahiri (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley and Sons

10.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) False. Refer to Section 10.3

2) $(\beta_2 + \beta_4), (\beta_3 + \beta_4), \beta_5, \beta_6$. Refer to Section 10.2

Check Your Progress 2

1) (a) True. Refer to Sub-Section 10.4.1, points 2 and 3.

   (b) False. Refer to Sub-Section 10.4.1, points 2 and 3.

Check Your Progress 3

1) (a) Refer to Section 10.5, point 7.

   (b) Refer to Section 10.5, point 8.

2) R^2 is high but the t-ratios are insignificant which is suggestive of multicollinearity.

3) True. Refer to Section 10.5, point 1 and 6. Some hints can also be obtained from Section 10.6.